

UPI Digital Payments Analytics — End-to-End Data Analyst Capstone

Project Overview

This project is an end-to-end analytics solution for a synthetic UPI/digital-payments platform. It moves from business understanding and data validation through SQL modelling, Python exploratory analysis, formal statistical testing, interactive Power BI reporting, drillthrough investigation, and executive recommendations.

The project was intentionally extended beyond the prescribed requirements to demonstrate the broader analytical workflow expected from a Data Analyst:

Business Question → Data Validation → SQL → Python/EDA → Statistical Testing → Power BI → Investigation → Business Action

Business Problem

A digital-payments platform needs visibility into:

- Transaction volume and monetary exposure
- Successful, failed, pending and reversed transactions
- Fraud activity and fraud exposure
- Customer, device, merchant, channel and regional risk patterns
- Operational failure patterns
- Segments that warrant deeper investigation

The analysis therefore focuses on both **performance** and **risk**, rather than treating fraud as an isolated KPI.

Analytical Objectives

1. Measure overall transaction and payment performance.
 2. Identify and quantify fraud exposure.
 3. Investigate device, customer, merchant, channel and regional risk patterns.
 4. Identify operational bottlenecks behind failed transactions.
 5. Test important hypotheses statistically.
 6. Build interactive dashboards that support investigation, not just reporting.
 7. Translate validated findings into practical business recommendations.
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Data

The project uses seven related datasets:

1. customer_master
2. device_info
3. upi_account_details
4. merchant_info
5. upi_transaction_history
6. customer_feedback_surveys
7. fraud_alert_history

The transaction history contains 100,000 transactions in the validated project dataset.

Tools & Technologies

Area	Tools
Business planning	PowerPoint
Data validation	Excel
Database	MySQL
Data analysis	Python, Pandas
Visualization	Matplotlib, Seaborn, Plotly
Statistical analysis	SciPy / statistical testing workflows
BI & reporting	Power BI
Interactive BI	Power BI slicers, drillthroughs and multi-page investigation views

Project Workflow

Step 1 — Business Understanding & Analytical Planning

Defined the business challenges, analytical objectives, KPI framework and hypotheses.

Deliverable: - UPI_Business_Understanding_Analytical_Planning.pptx

Step 2 — Excel Data Validation & Cleaning

Used the supplied workbook as a controlled validation layer. Data-quality checks included relationships, blanks, inconsistent categories, date/time normalization and validation helper columns.

Step 3 — SQL Database Design

Built the relational SQL structure across the seven datasets using primary keys, foreign keys, indexes and validation constraints where appropriate.

Step 4 — SQL Ingestion & Validation

Loaded and audited the datasets in MySQL, checking row counts, primary-key uniqueness, schema consistency and data types.

Step 5 — Python–SQL Integration

Established a reusable analytical workflow between Python and MySQL.

Step 6 — Python Cleaning, Transformations & EDA

Created analytical features and explored fraud, transaction, device, customer, merchant, channel, regional and temporal behaviour.

Step 7 — Statistical Analysis

Converted descriptive observations into formal hypothesis tests using:

- Chi-square tests
- Point-biserial correlation
- Welch independent-samples t-test
- One-way ANOVA
- Effect sizes
- Confidence intervals
- Assumption checks

Step 8 — Power BI Dashboard & Investigation

Built:

- Executive Command Center
- Fraud & Risk Intelligence
- Operations & Failure Intelligence
- Risk Investigation
- Region Drillthrough
- Device Drillthrough
- Merchant Drillthrough

A reusable Power BI measurement layer contains 20 DAX measures.

Step 9 — Strategic Insights & Executive Report

Translated the analysis into evidence-based findings, business implications and prioritized recommendations.

Core KPIs

The Power BI model includes measures for:

- Total Transactions
 - Total Transaction Value
 - Average Transaction Amount
 - Successful Transactions
 - Failed Transactions
 - Pending Transactions
 - Success Rate
 - Failure Rate
 - Fraud Transactions
 - Fraud Rate
 - Reversed Transactions
 - Reversal Rate
 - Rooted Transactions
 - Rooted Device Fraud
 - Rooted Device Fraud Rate
 - Fraud Transaction Value
 - Failed Transaction Value
 - Fraud Value Share
 - Transaction Value per Transaction
 - Rooted vs Non-Rooted Fraud Gap
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Hypothesis Framework

ID	Hypothesis	Test
H1	Rooted devices are associated with higher fraud occurrence.	Chi-square + Phi
H2	Fraud occurrence differs across transaction channels.	Chi-square + Cramér's V
H3	Device type is associated with transaction status.	Chi-square + Cramér's V
H4	Higher customer risk scores are associated with fraud occurrence.	Point-biserial correlation

ID	Hypothesis	Test
H5	Average transaction amount differs between Android and iOS users.	Welch t-test + 95% CI
H6	Mean merchant-level fraud rates differ across merchant types.	One-way ANOVA + eta-squared
H7	Mean customer-level fraud rates differ across regions.	One-way ANOVA + eta-squared

Key Analytical Findings

1. Rooted-device risk signal

Rooted-device transactions showed an observed fraud rate of approximately **20.69%**, compared with approximately **1.39%** for non-rooted-device transactions.

This is an observed difference of approximately **19.30 percentage points**.

The association was statistically significant in the project's chi-square analysis.

Important: this is an association/risk signal, not evidence that rooted status itself causes fraud.

2. Merchant-type ANOVA

Merchant-type analysis produced:

- $F = 0.618320$
- $p = 0.685911$
- eta-squared = **0.006219**

The tested data did not provide sufficient evidence of differences in mean merchant-level fraud rates across merchant types, and the estimated effect size was negligible.

This is an example of deliberately reporting a non-significant result rather than forcing a business narrative.

3. Statistical-method redesign

The regional ANOVA design was changed from aggregate region-level rates to **customer-level fraud rates** after a small-sample warning showed that the initial one-observation-per-region design was not appropriate.

This demonstrated methodological discipline: statistical warnings were treated as analytical signals rather than ignored.

What Makes This Project Stand Out

This project was intentionally developed beyond the prescribed requirements to demonstrate how a Data Analyst can move from raw business questions to deeper analytical discovery, advanced visualisation, interactive investigation and actionable business insight.

Rather than stopping at standard data cleaning and dashboarding, the project adds interactive 3D analytical landscapes, statistically validated risk discovery, effect sizes and confidence intervals, a multi-page Power BI investigation architecture, and dedicated Region, Device and Merchant drillthroughs. The strongest findings are tied back to business implications and monitoring actions, while non-significant results and methodological limitations are explicitly documented.

Standout Work Beyond the Brief

Interactive 3D Analytics

Created interactive Plotly-based 3D analytical landscapes to explore combinations of:

- Region
- Device
- Channel
- Merchant
- Failure behaviour
- Time-of-day

Investigation-oriented Power BI

Power BI was designed as an investigation environment rather than only a reporting layer.

Users can move from:

Signal → Segment → Drill Through → Investigate → Act

Drillthrough Architecture

Dedicated drillthrough paths were implemented for:

- Region
- Device
- Merchant

This allows users to move from an aggregate chart into a filtered detailed profile.

Multi-page Risk Workflow

The Power BI report separates:

- Executive monitoring
- Fraud/risk analysis
- Operations/failure analysis
- Detailed risk investigation

This keeps the executive view clean while retaining analytical depth.

Business Recommendations

The final recommendations are evidence-linked and include:

1. **Enhanced device-risk monitoring**
Consider stronger monitoring or step-up controls for transactions associated with rooted/jailbroken devices.
 2. **Risk-based segmentation**
Combine device, customer, channel, merchant, transaction-value and regional signals instead of relying on a single categorical factor.
 3. **Operational exposure monitoring**
Monitor failure rate together with failed transaction value to distinguish frequent issues from financially material issues.
 4. **Targeted investigation workflow**
Use interactive filters and drillthroughs to move from broad signals to detailed regional, device and merchant investigation.
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What Makes This Project Different

The project goes beyond a basic dashboard submission by combining:

Relational data modelling + data-quality validation + Python EDA + 3D analytics + formal statistics + effect sizes + confidence intervals + Power BI DAX + interactive drillthrough + executive recommendations

The objective was not merely to produce charts, but to demonstrate the reasoning and workflow expected from a Data Analyst.

Repository / Submission Structure

```
UPI_Digital_Payments_Analytics/  
  
  01_Business_Understanding/  
    UPI_Business_Understanding_Analytical_Planning.pptx  
  
  02_Data/  
    Raw_Data/  
      customer_master.csv  
      device_info.csv  
      upi_account_details.csv  
      merchant_info.csv  
      upi_transaction_history.csv  
      customer_feedback_surveys.csv  
      fraud_alert_history.csv  
  
    Validated_Data/  
      UPI_cleaned_KS.xlsx  
      Data Quality Log.csv  
  
  03_SQL/  
    UPI_Analytics_Schema.sql  
    UPI_Analytics_Analysis_Queries.sql  
  
  04_Python/  
    UPI_Transaction_Analysis.ipynb  
    3D_Visualizations/  
      [interactive 3D outputs / recordings]  
  
  05_Power_BI/  
    UPI_Transaction_Analytics.pbix  
  
  06_Executive_Report/  
    UPI_Executive_Report.pdf  
    UPI_Executive_Report.docx  
  
  07_Project_Documentation/  
    UPI_Project_Instructions.pdf  
  
  README.md
```

Important Notes

- The dataset is synthetic and should be described as such where appropriate.

- Statistical findings represent the tested dataset and should not automatically be generalized to real-world UPI populations.
 - Statistical association should not be described as causation.
 - KPI values can change under Power BI filter context.
 - Reproducibility depends on the included source data, SQL setup and Python environment.
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Project Outcome

This capstone demonstrates an end-to-end Data Analyst workflow:

Business understanding → data validation → relational modelling → analysis → statistical reasoning → advanced visualization → interactive BI → investigation → executive decision support